

BitMonie Development Update Report ?

Here is a comprehensive summary of the features and fixes we successfully implemented and deployed in our latest session. This covers the new savings modules, statement generation, and admin capabilities.

1. Savings Products Suite (Flex & SafeLock)

We have successfully engineered the core backend architecture for two new savings products, complete with automated daily interest accruals and strict business logic.

Flex Savings

- **Yield Model:** Implemented a **5% monthly yield** (~0.16% daily / 60% APR).
- **Accrual Engine:** Interest is calculated on the eligible balance and accrued daily via a background worker.
- **Withdrawal Logic:** 100% flexible. Users can withdraw their capital and any accrued profits at any time, any day, directly to their NGN wallet without penalty.

SafeLock (Fixed Savings)

- **Dynamic Yield Tiers:** Implemented fixed flat-rate returns based on lock durations:
 - 30 Days ? 6% Return
 - 60 Days ? 15% Return
 - 90 Days ? 20% Return
- **Maturity Logic:** Capital and accrued interest are securely locked. The system only credits the user's wallet automatically when the maturity date is reached.
- **Early Withdrawal (Break Lock) Logic:** If a user forcibly breaks a lock before maturity, they forfeit 100% of the accrued interest and incur a 1% administrative penalty on their principal.

2. Statement of Account Generation (PDF & CSV)

We built a robust, "fintech-grade" statement generation engine.

- **Formats Supported:** Users can export their transaction history in both **PDF** and **CSV** formats.
- **Asynchronous Delivery:** To ensure high performance, statement generation is offloaded to a background queue. Once generated, the statement is securely emailed to the user as an attachment.
- **PDF Design Upgrade:** Redesigned the PDF template to be highly professional.
 - Integrated the transparent BitMonie logo.
 - Added dynamic date ranges (e.g., `2024-03-01 to 2024-06-25` instead of generic fallbacks).
 - Resolved the `?` encoding bug by replacing the unsupported `?` glyph with a clean `NGN` designation.
 - Replaced generic footers with top-tier legal/security disclaimers.

3. Admin Panel & CRUD Enhancements

- **Savings Plans Management:** Built a complete Admin CRUD interface for SafeLock tiers. Admins can now dynamically create, edit, or toggle lock durations and interest rates directly from the dashboard under **Features -> Savings Plans** (No hardcoding required!).
- **User Dashboard Fixes:** Successfully investigated and resolved a bug on the active users table (`/admin/users/active`) where user phone numbers were failing to render.

4. Automation (Cron Jobs)

- Built and scheduled `ProcessDailyInterestJob` to run autonomously. This engine safely processes thousands of user wallets, calculates eligible interest days, applies the correct math based on product type, and securely updates ledger balances.

Status: All endpoints are fully operational, tested, and ready for frontend integration!